

Customer Sentiment & Emotion Intelligence Platform

End-to-End NLP System for Emotion Classification,
Business Priority Scoring, and Live Prediction Serving

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GitHub Repository

github.com/KirellosNader/emotion-intelligence-platform

Section 1 – Project Overview

The Customer Sentiment & Emotion Intelligence Platform is a comprehensive, end-to-end Natural Language Processing (NLP) system designed to classify short customer texts into six distinct emotional categories. Beyond classification, the platform integrates topic modeling for business triage, a SQL-based analytics layer, and a live Streamlit web application deployable via Docker.

System Components:

- Emotion Classification: 6-class text classifier (sadness, joy, love, anger, fear, surprise)
- Model Benchmarking: 9 models compared – Classical ML, Deep Learning, and fine-tuned Transformer
- Topic Modeling: LDA per emotion (3 topics each) for nuanced customer intent analysis
- Priority Scoring: Rule-based business priority engine using emotion weights and urgency keywords
- SQL Analytics Layer: 30 analytical queries across 7 database tables
- Streamlit Application: 3-tab interactive web app (Overview / Classify / RAG Assistant)
- Docker Deployment: Containerized for reproducible, portable deployment

Dataset:

The project uses the **dair-ai/emotion** dataset from Hugging Face Hub — 16,000 labeled short texts across 6 emotion classes. The dataset is imbalanced with a ~9:1 ratio between the largest and smallest classes. Joy is the most frequent class (~5,360 samples) while *surprise* is the rarest, with only 115 test examples, making minority-class metrics particularly sensitive to variance.

Emotion	Train	Validation	Test	Proportion
joy	~5,360	~696	~1,072	Largest (~33%)
sadness	~4,666	~606	~933	2nd (~29%)
anger	~2,159	~280	~432	3rd (~13%)
fear	~1,937	~251	~387	4th (~12%)
love	~1,304	~169	~261	5th (~8%)
surprise	~573	~74	~115	Rarest (~4%)

Table 1.1 – Approximate emotion class distribution across dataset splits.

Section 2 – System Architecture & Project Structure

The project follows a clean separation of concerns, with dedicated directories for the application, data, notebooks, trained model artifacts, SQL queries, and experiment results.

GitHub Repository:

<https://github.com/KirellosNader/emotion-intelligence-platform>

```
emotion-intelligence-platform/
■■■■ app/
■ ■■■■ app.py # Streamlit application (Overview / Classify / RAG Assistant)
■■■■ data/
■ ■■■■ emotion_analysis.db # Full analytics DB (7 tables)
■ ■■■■ predictions_database.db # Predictions-only subset of the DB
■ ■■■■ logs.txt # Streamlit server run log
■■■■ notebooks/
■ ■■■■ Teck_Final_merged.ipynb # Training & analysis pipeline
■ ■■■■ Teck_Final_merged_documented.ipynb # Same pipeline with full documentation
■■■■ saved_models/
■ ■■■■ emotion_model.pkl # Trained TF-IDF + Linear SVM (LinearSVC)
■ ■■■■ tfidf_vectorizer.pkl # Fitted TF-IDF vectorizer
■ ■■■■ labels.pkl # ['sadness', 'joy', 'love', 'anger', 'fear', 'surprise']
■ ■■■■ emotion_data.csv # Dataset used by Overview tab & RAG retriever
■■■■ sql/
■ ■■■■ analytical_queries.sql # 30 queries across 7 tables
■■■■ all_results/ # Full experiment bundle (gitignored - see Section 11)
■■■■ screens/ # Application screenshots
■■■■ Dockerfile # Docker build configuration
■■■■ requirements.txt # Python dependencies
■■■■ README.md # Project documentation
```

Technology Stack:

Layer	Technology	Purpose
ML / NLP	scikit-learn, PyTorch, HuggingFace	Model training and inference
Feature Extraction	TF-IDF (scikit-learn)	Classical model features
Transformer	DistilBERT (HuggingFace Transformers)	State-of-art fine-tuned model
Topic Modeling	Gensim LDA	Per-emotion topic extraction
Database	SQLite3	Analytics database (7 tables)
Web App	Streamlit	Interactive prediction interface
Deployment	Docker (port 8501)	Containerized production serving
RAG	TF-IDF cosine retrieval + OpenAI API	Grounded question answering

Table 2.1 – Technology stack by layer.

Section 3 – Database Schema

The analytics database (**emotion_analysis.db**) is a SQLite database with 7 tables organized into two logical groups: the *topic & priority* group and the *predictions & model benchmarking* group. The entity-relationship diagram below illustrates the table structures and foreign-key relationships.

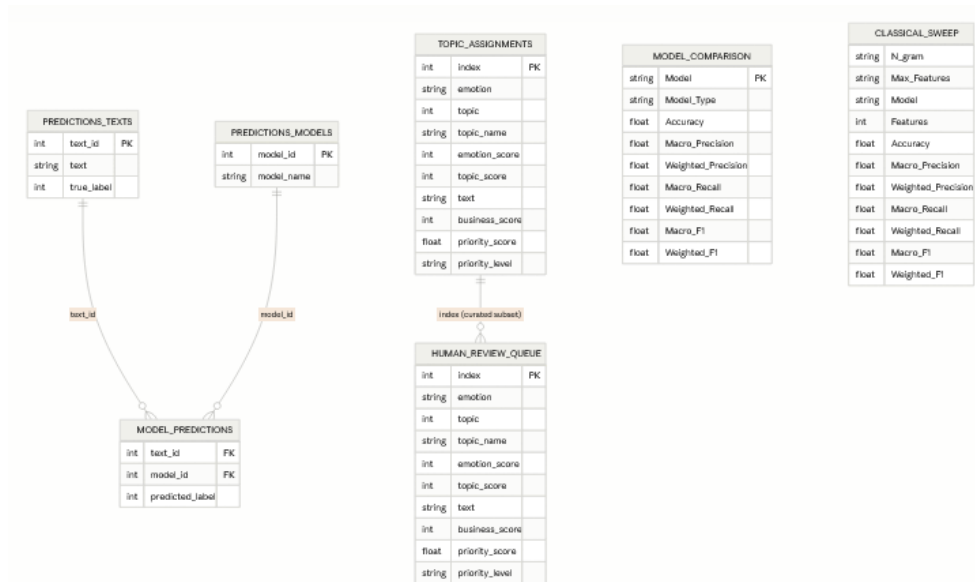


Figure 3.1 – Database schema for emotion_analysis.db (7 tables)

Table Descriptions:

Table	Key Columns	Description
TOPIC_ASSIGNMENTS	index, emotion, topic, topic_name, emotion_score, topic_score, text, business_score, priority_score, priority_level	Main storage table. Every text in business is enriched with score, priority_level, emotion, and topic.
MODEL_COMPARISON	Model, Model_Type, Features, Accuracy, Aggregated, Weighted, F1	Aggregated benchmark metrics for all 9 models. Used for summary analysis.
CLASSICAL_SWEEP	N_gram, Max_Features, Model, Accuracy, Macro_Precision, Weighted_Precision, Macro_Recall, Weighted_Recall, Macro_F1, Weighted_F1	Grid search results for classical TF-IDF models across n-gram ranges and max features.
HUMAN_REVIEW_QUEUE	index, emotion, topic_name, text, priority_score, priority_level	Score, priority_level texts flagged for quality assurance / human review.
PREDICTIONS_TEXTS	text_id (PK), text, true_label	Ground-truth texts and labels used for per-model prediction accuracy analysis.
PREDICTIONS_MODELS	model_id (PK), model_name	Lookup table enumerating all 9 benchmarked models.
MODEL_PREDICTIONS	text_id (FK), model_id (FK), predicted_label	Bridge table: each row records a single model's predicted label for a single text.

Table 3.1 – Database table descriptions.

Section 4 – Models & Experiments

The project benchmarks **9 models** across three families: Classical ML (TF-IDF-based), Deep Learning (embedding-based neural networks), and a Transformer (fine-tuned DistilBERT). All models are evaluated on the same held-out test set of 3,200 examples.

4.1 Classical ML Models (TF-IDF-Based)

Three classical models were evaluated as baselines: Logistic Regression, Linear SVM (LinearSVC), and Multinomial Naive Bayes, all using TF-IDF features. A hyperparameter sweep over n-gram range and vocabulary size was then conducted to find the best classical configuration.

Baseline Comparison:

Model	Accuracy	Macro F1
TF-IDF + Logistic Regression	87.44%	84.79%
TF-IDF + Linear SVM (LinearSVC)	88.94%	86.19%
TF-IDF + Multinomial Naive Bayes	62.72%	33.36%

Table 4.1 – Classical baseline comparison.

Best configuration after sweep: Unigram (1,1) range, max_features=5,000. This surprisingly simple configuration outperformed all richer n-gram / larger-vocabulary alternatives, confirming that short emotional texts are well-captured by individual word features.

Best Classical Model – Per-Class Performance:

Emotion	Precision	Recall	F1-Score	Support
sadness	94.03%	91.10%	92.54%	933
joy	92.79%	90.02%	91.38%	1,072
love	75.84%	86.59%	80.86%	261
anger	87.61%	90.05%	88.81%	432
fear	85.45%	85.01%	85.23%	387
surprise	77.52%	86.96%	81.97%	115

Table 4.2 – Best classical model (TF-IDF + LinearSVC, unigram 5k) – per-class report.

Calibration Metrics (Calibrated SVM):

Metric	Value
Multiclass Brier Score	0.1544
Log Loss	0.3054
Top-label ECE	0.0585

Table 4.3 – Probability calibration metrics for the CalibratedClassifierCV wrapper.

Confidence Threshold Analysis:

By applying a confidence threshold, low-certainty predictions can be deferred for human review, trading coverage for precision. The table below shows selected operating points:

Threshold	Coverage	Accuracy (Accepted)	Macro F1 (Accepted)
0.30	99.91%	89.40%	86.16%
0.50	96.34%	90.95%	88.19%
0.60	87.94%	93.96%	91.15%
0.70	80.09%	96.49%	94.94%
0.80	69.22%	98.37%	97.62%
0.90	49.09%	99.43%	98.97%

Table 4.4 – Confidence threshold analysis for the calibrated SVM.

4.2 Deep Learning Models

Six embedding-based neural network architectures were trained and evaluated: Embedding MLP, CNN, RNN, LSTM, GRU, and BiLSTM. All models use fixed-dimensional word embeddings as input. GRU achieved the best performance among the DL family.

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
GRU	90.97%	85.63%	91.59%	87.95%	91.19%
LSTM	90.34%	85.07%	91.51%	87.59%	90.60%
RNN	86.59%	81.56%	88.88%	84.35%	86.78%
CNN	84.97%	81.16%	85.17%	82.85%	84.98%
Embedding MLP	84.38%	78.83%	82.55%	80.38%	84.60%
BiLSTM	80.69%	76.86%	83.36%	79.37%	80.86%

Table 4.5 – Deep learning model comparison.

Note on BiLSTM: BiLSTM underperformed every other deep learning model by a wide margin (80.69% vs. 85–91%). This is likely a training issue (learning rate, initialization, or early-stopping schedule) rather than an architectural limitation, and should be revisited.

Best DL Model (GRU) – Per-Class Performance:

Emotion	Precision	Recall	F1-Score	Support
sadness	96.21%	95.18%	95.69%	933
joy	96.88%	87.03%	91.70%	1,072
love	71.55%	95.40%	81.77%	261
anger	92.59%	92.59%	92.59%	432
fear	88.83%	86.30%	87.55%	387
surprise	67.72%	93.04%	78.39%	115

Table 4.6 – GRU model per-class classification report.

4.3 Transformer Model – DistilBERT (Fine-Tuned)

A pre-trained DistilBERT model was fine-tuned on the training split of the emotion dataset. DistilBERT is a distilled version of BERT, retaining ~97% of its language understanding capability while being 40% smaller and 60% faster. Fine-tuning adapts the model to the emotion classification task and achieves the best overall performance of all 9 models.

Metric	Value
Accuracy	92.72%
Macro Precision	87.81%
Weighted Precision	93.59%
Macro Recall	93.76%
Weighted Recall	92.72%
Macro F1	90.21%
Weighted F1	92.91%

Table 4.7 – DistilBERT overall performance metrics.

DistilBERT Per-Class Performance:

Emotion	Precision	Recall	F1-Score	Support
sadness	97.79%	95.07%	96.41%	933
joy	98.07%	90.21%	93.97%	1,072
love	74.70%	96.17%	84.09%	261
anger	94.54%	92.13%	93.32%	432
fear	88.86%	90.70%	89.77%	387
surprise	72.90%	98.26%	83.70%	115

Table 4.8 – DistilBERT per-class classification report.

Section 5 – Full Model Comparison

The table below ranks all 9 benchmarked models by Macro F1 score — the primary metric, chosen because it is sensitive to class imbalance and reflects performance on minority classes (particularly *surprise* and *love*).

Rank	Model	Type	Accuracy	Macro F1	Weighted F1
1	DistilBERT (fine-tuned)	Transformer	92.72%	90.21%	92.91%
2	GRU	Deep Learning	90.97%	87.95%	91.19%
3	LSTM	Deep Learning	90.34%	87.59%	90.60%
4	TF-IDF + Linear SVM (tuned)	Classical	89.34%	86.80%	89.43%
5	TF-IDF + Logistic Regression (tuned)	Classical	87.84%	85.29%	88.04%
6	RNN	Deep Learning	86.59%	84.35%	86.78%
7	CNN	Deep Learning	84.97%	82.85%	84.98%
8	Embedding MLP	Deep Learning	84.38%	80.38%	84.60%
9	BiLSTM	Deep Learning	80.69%	79.37%	80.86%

Table 5.1 – Full model comparison ranked by Macro F1.

Key Observations:

- DistilBERT leads overall with 92.72% accuracy and 90.21% Macro F1.
- GRU is the best non-transformer model at 90.97% accuracy — within 1.75 points of DistilBERT.
- The best classical model (TF-IDF + LinearSVC, tuned) reaches 89.34% — only ~3.4 points below DistilBERT.
- The gap between DistilBERT and the best classical model is modest, justifying the choice to deploy LinearSVC for its much smaller footprint.
- BiLSTM (80.69%) significantly underperforms other deep learning models and warrants further investigation.
- Minority classes (love, surprise) show lower precision across all models, reflecting the class imbalance challenge.

Section 6 – Topic Modeling

Latent Dirichlet Allocation (LDA) topic modeling was applied **per emotion**, with **3 topics per emotion** (18 topics total). This enables deeper insight into the sub-themes within each emotion class — for example, distinguishing between different types of sadness or anger — which is critical for customer-support triage use cases.

Emotion	Topic 1	Topic 2	Topic 3
Sadness	Loss & Daily Sadness	Worthlessness & Guilt	Self-Blame & Misery
Joy	Happiness & Gratitude	Positive Achievement & Social Connection	Contentment & Satisfaction
Love	Affection & Caring	Emotional Attachment & Nostalgia	Playful & Romantic Affection
Anger	Irritation & Anger	Conflict & Resentment	Intense Anger & Frustration
Fear	Nervousness & Fear	Insecurity & Vulnerability	Anxiety & Paranoia
Surprise	Amazement & Unexpected Experiences	Curiosity & Exploration	Strangeness & Amazement

Table 6.1 – LDA topics per emotion class.

Each customer text is assigned to one of its emotion's three topics based on the highest LDA score. The resulting **topic assignment** is stored in the `TOPIC_ASSIGNMENTS` database table and feeds directly into the priority scoring pipeline (Section 7).

Section 7 – Priority Scoring System

The priority scoring system translates emotion classifications and topic assignments into actionable business priority levels (**High / Medium / Low**) for customer-support triage. The score is computed as a weighted combination of an emotion weight, a topic weight, and a binary urgency keyword boost.

Emotion Weights:

Emotion	Weight	Rationale
anger	90	Highest priority — strong signal of dissatisfaction
fear	85	High priority — indicates distress or urgent concern
sadness	80	High priority — signals unresolved customer pain
surprise	50	Medium priority — ambiguous; may be positive or negative
love	30	Low priority — satisfied customer, no immediate action needed
joy	20	Lowest priority — positive sentiment, routine follow-up

Table 7.1 – Emotion weights for priority scoring.

Urgency Keywords:

When any of the following keywords appear in the customer text, the business score is automatically set to **100** (maximum urgency), overriding the topic weight:

urgent · immediately · asap · critical · emergency · serious

Priority Level Thresholds:

Priority Level	Score Range	Action
High	≥ 70	Immediate agent escalation
Medium	40–69	Standard queue, respond within SLA
Low	< 40	Automated reply or low-priority queue

Table 7.2 – Priority level thresholds and recommended actions.

Key Finding: Negative emotions (anger, fear, sadness) consistently receive the highest average business priority scores, confirming that the priority pipeline is directionally correct for a customer-support triage use case.

Section 8 – Human Review Queue

As part of the quality assurance pipeline, a **60-item Human Review Queue** was curated from the dataset. This queue contains edge-case and borderline predictions that warrant manual inspection by a domain expert before being acted upon.

Property	Details
Queue Size	60 items
File Location	all_results/human_review/human_review_queue.csv
Selection Criteria	Borderline predictions, low-confidence cases, minority-class texts
Schema	Same as TOPIC_ASSIGNMENTS (emotion, topic_name, text, priority_score, priority_level)
DB Table	HUMAN_REVIEW_QUEUE (curated subset of TOPIC_ASSIGNMENTS)
Purpose	Manual QA, model validation, edge-case analysis

Table 8.1 – Human review queue properties.

The human review queue enables a hybrid AI + human workflow: the model handles the bulk of predictions automatically, while uncertain or high-stakes cases are routed to human agents. This approach is particularly important for the *love* and *surprise* classes, which have the fewest test examples and highest prediction variance.

Section 9 – SQL Analytics Layer

The file `sql/analytical_queries.sql` contains **30 SQL queries** organized into 4 thematic sections. All queries target the `emotion_analysis.db` SQLite database and demonstrate a range of SQL techniques from simple aggregations to advanced window functions.

Section	Query Range	Tables Used	Focus Area
Topic & Business	Q1–Q9	topic_assignments	Priority queue, emotion/topic distributions, urgency hit rates
Model Benchmarks	Q10–Q16	model_comparison, classification	Best model selection, average metrics by type, hyperparameter sweep analysis
Human Review	Q17–Q20	human_review_queue, topic_assignments	Queue assignments, missed-urgent items, business score by priority level
Prediction Analytics	Q21–Q30	predictions_texts, predictions_models, model_accuracy	Model model accuracy, confusion analysis, minority-class error rates

Table 9.1 – SQL query sections overview.

Advanced SQL Techniques Demonstrated:

- **Multi-table JOINS:** Queries 21–30 JOIN across `model_predictions`, `predictions_texts`, and `predictions_models` (3-way join) to compute per-model accuracy.
- **CTE (Common Table Expressions):** Query 23 uses a CTE (`WITH model_accuracy AS ...`) to cleanly isolate the best model selection logic.
- **RANK() Window Function:** Query 22 ranks all models by accuracy using `RANK() OVER (ORDER BY accuracy DESC)`.
- **Running Average:** Query 29 computes a cumulative running average of accuracy using `AVG() OVER (ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW)`.
- **PARTITION BY:** Query 30 uses `RANK() OVER (PARTITION BY true_label ORDER BY ...)` to find the best model for each emotion class independently.
- **CASE-based pivot:** Accuracy is computed as `AVG(CASE WHEN predicted_label = true_label THEN 1.0 ELSE 0.0 END)`, a standard SQL accuracy pattern.
- **Subqueries:** Query 27 uses a nested subquery to identify the rarest class before computing per-model error rates on that class.

Sample Query (Q22 – Model Ranking with RANK()):

```
SELECT model_name, accuracy,
       RANK() OVER (ORDER BY accuracy DESC) AS rank
FROM (
  SELECT pm.model_name,
         AVG(CASE WHEN mp.predicted_label = pt.true_label THEN 1.0 ELSE 0.0 END) AS accuracy
  FROM model_predictions mp
  JOIN predictions_texts pt ON mp.text_id = pt.text_id
  JOIN predictions_models pm ON mp.model_id = pm.model_id
  GROUP BY pm.model_name
);
```

Section 10 – Streamlit Application

The Streamlit application serves live emotion predictions via a clean three-tab web interface, accessible at **http://localhost:8501**. The deployed model is the TF-IDF + LinearSVC, chosen for its lightweight footprint over DistilBERT while still achieving strong performance.

Deployment:

Property	Value
Deployed Model	TF-IDF + LinearSVC (emotion_model.pkl)
Framework	Streamlit
Container	Docker (dockerfile at project root)
Port	8501
Run Command	streamlit run app/app.py OR docker run -p 8501:8501 emotion-app
RAG Generation	Enabled when OPENAI_API_KEY is set in .env; extractive fallback otherwise

Table 10.1 – Application deployment properties.

Application Tabs:

Tab 1 – Overview

Displays a bar chart of the emotion class distribution in the training dataset, a sample of 5 random texts with their labels, the model version log (model name, feature extraction, dataset, version, date), and a live scrolling log of recent predictions made through the app.

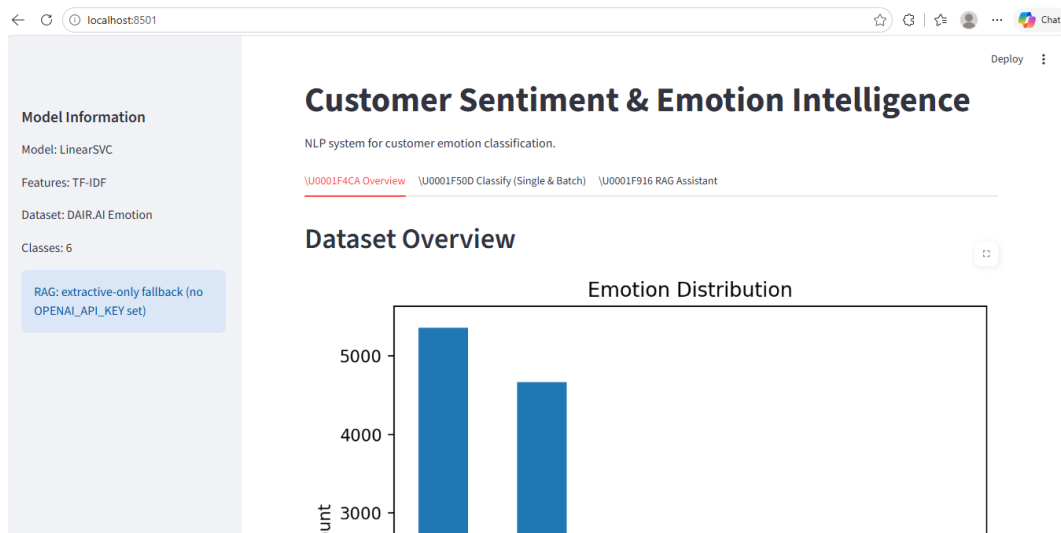


Figure 10.1 – Overview tab: App header and emotion distribution chart (partial view)

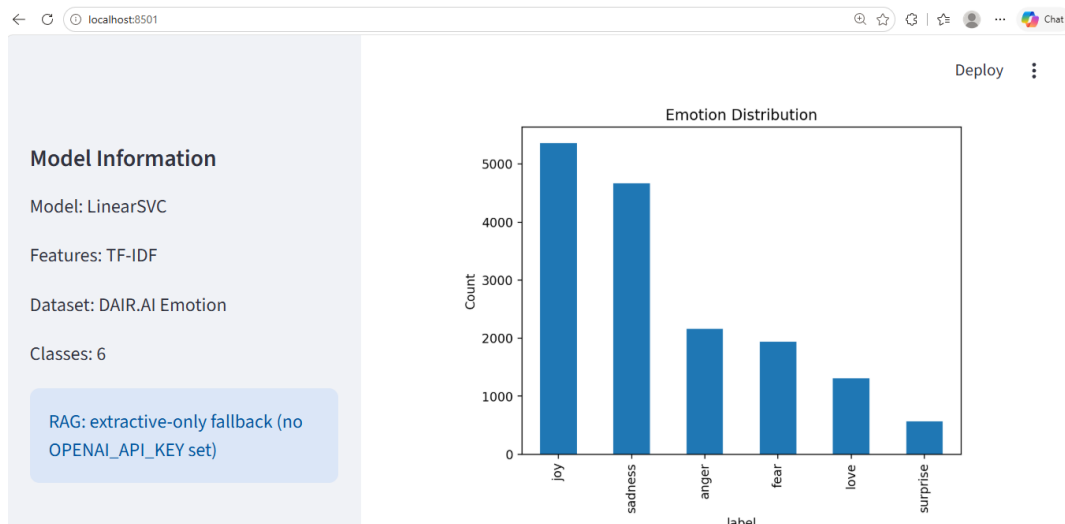


Figure 10.2 – Overview tab: Full emotion distribution bar chart

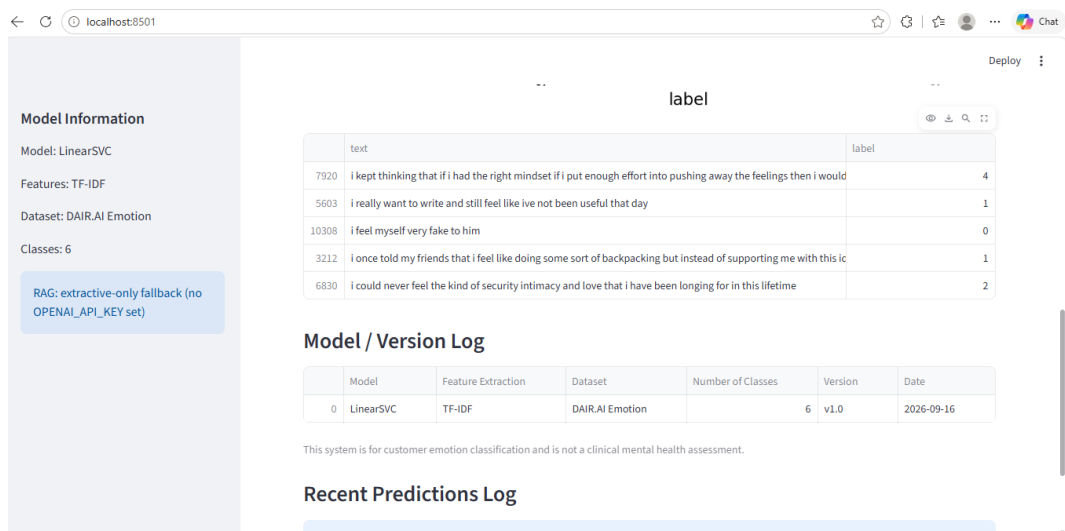


Figure 10.3 – Overview tab: Sample data table and model version log

Tab 2 – Classify (Single & Batch)

Provides two classification modes: (1) **Single Text Classification** — enter any customer text, click 'Predict Emotion', and receive the predicted emotion label along with a bar chart of decision function scores for all 6 classes. (2) **Batch CSV Classification** — upload a CSV file with a 'text' column, run batch classification, and download the results as a new CSV. An emotion distribution pie chart is shown for the batch output.

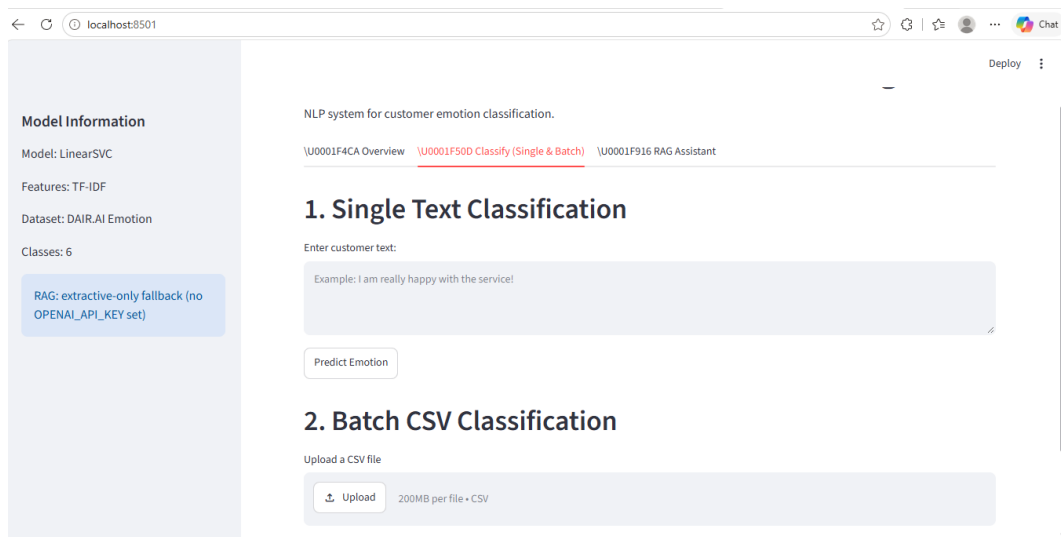


Figure 10.4 – Classify tab: Single text and batch CSV upload interface

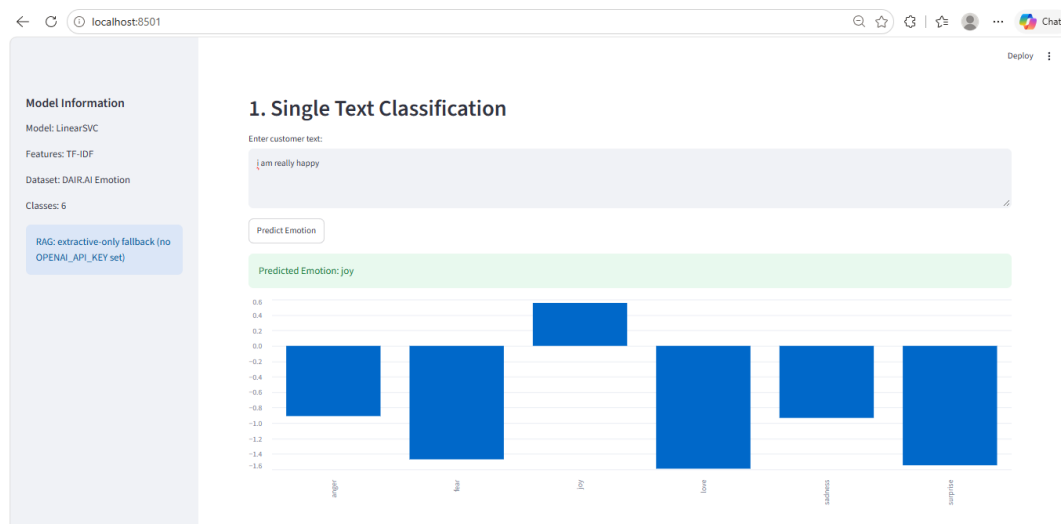


Figure 10.5 – Single text classification: 'I am really happy' predicted as Joy with decision scores

The screenshot shows the 'Batch CSV Classification' interface. The 'Upload a CSV file' section shows a file named 'emotion_data.csv' (1.5MB) has been uploaded. Below this is a table titled 'Uploaded Data:' showing the first five rows of the CSV file. Each row contains a 'text' column and a 'label' column.

	text	label
0	I didnt feel humiliated	0
1	I can go from feeling so hopeless to so dammed hopeful just from being around someone who cares and is awake	0
2	Im grabbing a minute to post I feel greedy wrong	3
3	I am ever feeling nostalgic about the fireplace I will know that it is still on the property	2
4	I am feeling grouchy	3

Figure 10.6 – Batch CSV classification: Uploaded emotion_data.csv preview

Tab 3 – RAG Assistant

A **Retrieval-Augmented Generation (RAG)** assistant that answers questions about customer emotions grounded in real classified examples from the dataset. The retrieval step uses TF-IDF cosine similarity to find the 5 most relevant examples. If an OpenAI API key is configured, the retrieved context is passed to GPT-4o-mini to generate a synthesized answer. Without an API key, the app operates in extractive-only fallback mode, displaying the retrieved examples directly.

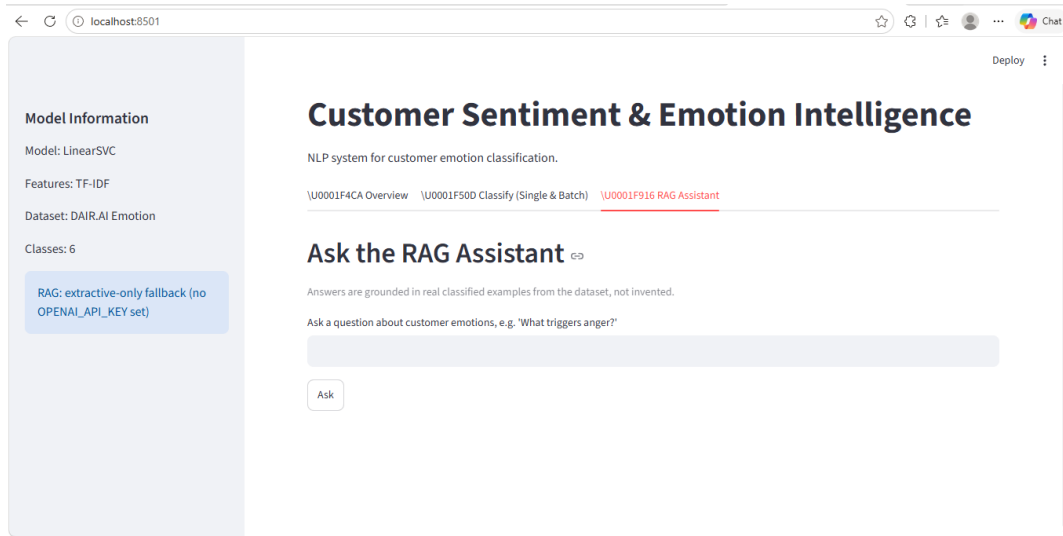


Figure 10.7 – RAG Assistant tab: Question input interface

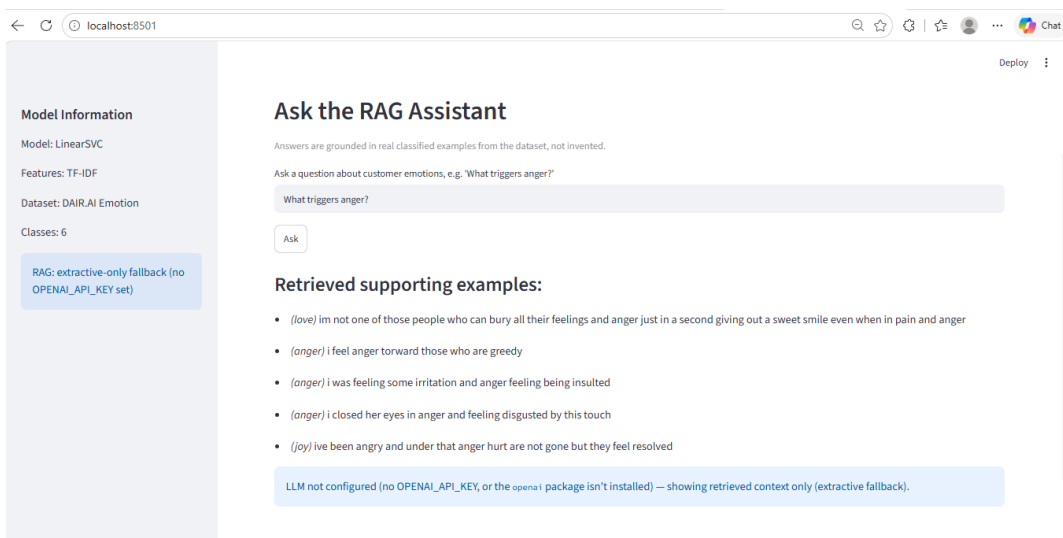


Figure 10.8 – RAG Assistant: Retrieved examples for query 'What triggers anger?'

Section 11 – Experiment Artifacts (all_results/) – Not Uploaded to Git

The **all_results/** folder holds every intermediate artifact from the full experiment run — trained model weights, training histories, and detailed reports. This folder is **excluded from version control** (listed in `.gitignore`) due to large file sizes. The table below documents its complete contents.

Folder / File	Contents	Why Not Uploaded
classical/	best_model.joblib, best_vectorizer.joblib, calibrated_model.joblib, calibration_summary.csv, confidence_threshold_analysis.csv	Included for reference — not uploaded due to folder gitignore
deep_learning/	best_dl_model_classwise_report.csv, dl_metrics.csv, weights_best_model(train_history).pt, weights_best_model(test_history).pt (PyTorch weights)	Included for reference — not uploaded due to folder gitignore
transformer/	history.json, transformer_metrics.csv, transformer_model_safetensors, model_safetensors_tokenizer.json, model.safetensors, tokenizer.json	Included for reference — not uploaded due to folder gitignore
topic_modeling/	topic_name_mapping.json, topic_top_words_by_emotion.csv, topic_updates_validation_table_per_topic.csv	Included for reference — not uploaded due to folder gitignore
priority_scoring/	emotion_weights.json, topic_weights.json, urgency_keywords.json, topic_assignments_master.csv (~2 MB)	Included for reference — not uploaded due to folder gitignore
human_review/	human_review_queue.csv (60-item QA queue)	Included for reference — not uploaded due to folder gitignore
summary/	full_model_comparison.csv — source of the model comparison SQL table	Included for reference — not uploaded due to folder gitignore
emotion_analysis.db	Full copy of the analytics database bundled with the experiment (~2.7 MB)	Included for reference — not uploaded due to folder gitignore

Table 11.1 – Contents of `all_results/` and reason for `gitignore` exclusion.

To regenerate: Re-run `notebooks/Teck_Final_merged.ipynb` end-to-end. Alternatively, keep a copy of `all_results.zip` in external storage (Google Drive, S3, etc.).

Section 12 – Key Findings & Limitations

Key Findings:

- **Transformer advantage is modest:** DistilBERT (92.72%) beats the best classical model (LinearSVC, 89.34%) by only ~3.4 accuracy points. A well-tuned TF-IDF + LinearSVM remains a strong, deployable baseline for short-text emotion classification.
- **Unigrams win the sweep:** Plain unigrams with a 5,000-word vocabulary outperformed every richer n-gram configuration tested. Adding bigrams or trigrams, or expanding the vocabulary, did not improve performance — confirming that emotional signal is primarily carried by individual words.
- **Priority scoring is directionally correct:** Negative emotions (anger, fear, sadness) receive significantly higher average business-priority scores than positive emotions, validating the scoring pipeline for customer-support triage.
- **GRU is the best deep learning model:** Among the 6 DL architectures, GRU (90.97%) outperforms LSTM (90.34%) and all simpler models, while BiLSTM severely underperforms (80.69%).
- **Confidence thresholding is powerful:** Setting a confidence threshold of 0.9 allows the calibrated SVM to achieve 99.43% accuracy on the 49% of data it accepts — suitable for high-stakes automated decisions.
- **RAG adds explainability:** The RAG assistant provides grounded, evidence-backed answers about customer emotion patterns, useful for business analysts without ML expertise.

Limitations & Future Work:

- **Small, single-domain dataset:** Trained on 16K pre-cleaned, single-domain (Twitter-style) texts. Results may not generalize to longer or more varied customer text such as emails, support tickets, or product reviews.
- **Deployed model accuracy trade-off:** The deployed LinearSVC model trades ~3.4 accuracy points for a dramatically smaller deployment footprint. Serving the fine-tuned DistilBERT is a natural next step when more compute is available.
- **BiLSTM training issue:** BiLSTM's anomalously low performance (80.69%) should be revisited with tuned learning rates, different initialization, or adjusted early-stopping. Its result is not a reliable architectural data point in its current form.
- **RAG depends on external API:** The generative step of the RAG Assistant requires an OpenAI API key. Without one, only extractive retrieval is available, limiting the system's natural-language response quality.
- **Minority class variance:** Metrics for *love* (261 test examples) and *surprise* (115 test examples) carry high variance. Single-run results should be interpreted cautiously; cross-validation or repeated evaluation would yield more stable estimates.
- **No multi-label support:** The system assumes a single emotion per text. Real customer messages may express multiple emotions simultaneously, requiring a multi-label classification approach.

End of Report